



TECHNICAL DESIGN ARCHITECTURE & RESOLUTIONS

1. Database Choice: Storage & Geospatial Scaling

- **Resolution:** Leverage **Supabase PostgreSQL** as the core relational database.
- **Blueprint Specification:** Rather than implementing a secondary system like Turso, we will utilize the native PostgreSQL relational instance provided in the Supabase Free Tier. It accommodates up to 500 MB of pure text data, which is more than sufficient for a catalog of over 2,000 grocery SKUs and active order logs. The developer will enable the built-in **PostGIS** extension via the dashboard with a single click, providing future-proof radius-based search parameters (e.g., finding local delivery zones within 5km) without requiring database migrations.

2. Payment Gateway: Transaction Framework

- **Resolution:** Standard API Integration via **Stripe Checkout**.
- **Blueprint Specification:** Exclude dedicated Direct Debit platforms like GoCardless from the initial MVP to reduce deployment complexity. Stripe Checkout operates with a **₹0/month fixed subscription fee**, billing exclusively on a pay-as-you-go transaction percentage. To manage potential future B2B or recurring grocery subscriptions, **Stripe Billing** can be activated directly within the same dashboard, enabling Direct Debit pathways under a single payment infrastructure.

3. Mobile Framework: Runtime Environment

- **Resolution:** Cross-Platform Native Compilation via **React Native Expo**.
- **Blueprint Specification:** Avoid hybrid wrapper frameworks like CapacitorJS. Since grocery apps require highly responsive catalog browsing and steady background data streaming (for rider location updates), React Native Expo is selected. This framework provides native UI rendering components, smoother gestures, and resilient background processing threads—all built on an open-source framework that costs nothing to develop.

4. Delivery Pricing: Revenue & Operations Model

- **Resolution:** Dynamic Variable Logic via an Administrative **Threshold Model**.
- **Blueprint Specification:** Code a flexible delivery logic within the backend database schema using three core parameters: **base_delivery_fee**, **per_km_charge**, and **free_delivery_threshold**. For the grocery store framework, we will default to a threshold model (e.g., *"Free delivery on orders above £20, otherwise a fixed delivery fee"*). This design structure incentivizes users to increase their cart value while giving the

store admin the power to change pricing configurations instantly from the dashboard without redeploying code.

5. Single vs. Multi-Store Architecture

- **Resolution:** Multi-Store Schema Compatibility with Single-Store Frontend Isolation.
- **Blueprint Specification:** The user interface will be strictly confined to a single grocery store setup to accelerate development speeds. However, the database schema will include a mandatory `store_id` relational key across all orders, products, and user profiles. This structural approach ensures that if the business expands to multi-branch operations, the core infrastructure remains unbroken, and additional store entities can be added seamlessly.

6. Timeline Risk: Scope Optimization

- **Resolution:** Integration of **Supabase Built-in Authentication**.
- **Blueprint Specification:** Building custom authentication models from scratch risks overextending the 8-week delivery timeline. To eliminate this risk, developers will implement **Supabase Auth** for handling user signups, passwords, and OTP management out-of-the-box. This cuts development overhead by roughly 2 weeks, ensuring the remaining lifecycle tracking and core grocery shopping modules are deployed on schedule.

7. HFSS Advertising Regulations (UK Legal Compliance)

- **Resolution:** Regulatory Compliance Parameterization via a Schema Boolean Flag.
- **Blueprint Specification:** Due to legislative guidelines restricting the promotion of High Fat, Sugar, and Salt (HFSS) items, a strict compliance engine is required for grocery catalogs. We will add an `is_hfss: boolean (default: false)` flag to the `products` database table. The frontend application logic will check this variable and automatically exclude any marked item from being displayed on promotional banners or top-tier discount spots, shielding the brand from legal compliance issues.

8. Right-to-Work Checks: Courier Onboarding Workflow

- **Resolution:** Manual Administrative Verification Engine via Secure Asset Buckets.
- **Blueprint Specification:** To comply with regional courier labor regulations, the rider onboarding module will require document photo uploads (Visas, IDs, Licenses) routed straight to a secure private folder inside **Supabase Storage**. A state parameter field `verification_status` with values (`'pending'`, `'approved'`, `'rejected'`) will block riders from accessing the active order matching pool until the store administrator manually verifies the credentials via the dashboard, keeping automation costs at ₹0.

IN SHORT

- 1. Database Choice:** Use the free tier of **Supabase PostgreSQL** with **PostGIS** turned on to get advanced location features for ₹0.
- 2. Payment Gateway:** Use standard **Stripe Checkout**, which has no monthly fees and charges you only per successful grocery order.
- 3. Mobile Framework:** Choose **React Native Expo** because it is free and gives a completely smooth, premium native app experience.
- 4. Delivery Pricing:** Code a **Flexible Threshold Model** so you can change fees or offer free delivery on large orders from your admin panel.
- 5. Single vs Multi-Store:** Keep the screen single-store for now, but add a hidden **store_id** column in the database to allow easy future scaling.
- 6. Timeline Risk:** Save 2 weeks of development time by using **Supabase Auth** for free, ready-made user logins and OTP structures.
- 7. HFSS Advertising:** Add a simple **is_hfss** True/False checkbox to your products so unhealthy items are automatically filtered out of promo spots.
- 8. Right-to-Work Checks:** Create a manual **Document Upload Status** field where riders submit IDs to a secure free folder for your manual approval.